

WASP-50b

R-1668

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_131217 · created 2026-08-02T10:18:36+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456643.739 +/- 0.09 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.05
Transit depth (Rp/Rs)^2	2.74 +/- 1.65 [%]
Orbital Inclination (inc)	89.05 +/- 2.92
Airmass coefficient 1 (a1)	0.282 +/- 0.098
Airmass coefficient 2 (a2)	0.83 +/- 0.9
Scatter in the residuals of the lightcurve fit is	31.14 %
Best Comparison Star	#1 - [87, 407]
Optimal Aperture	4.64
Optimal Annulus	5.7
Transit Duration (day)	0.052 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.05 vs 0.1404 (deviation 0.49σ)
Duration fit vs published	0.052 d vs 0.0754 d (deviation 0.63σ)
Mid-time O–C	71.7 min
Depth SNR	3.3
Residual scatter	31.14 %

Light curve

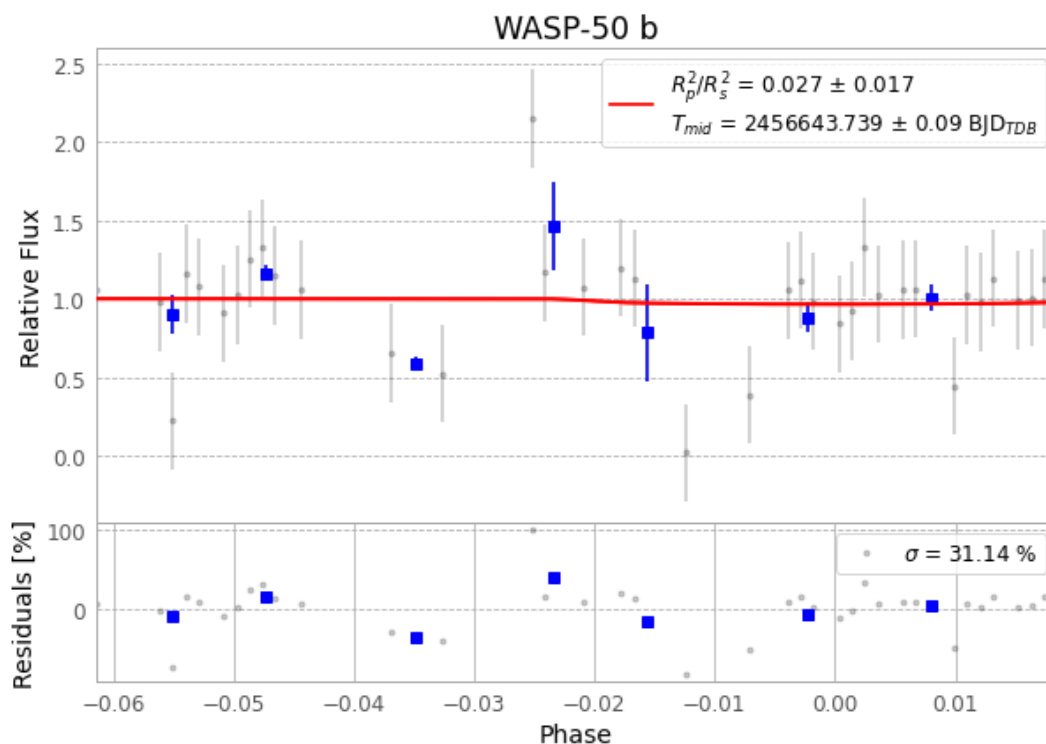


Figure 1 — FinalLightCurve_WASP-50 b_2013-12-16.png (SHA-256 b87d6f9f2e965a21...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [523, 216], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 53ec6ad8ef31a9a3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-50131217024512.FITS → WASP-50131217063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-131217.log (a156e366f33b...), FinalLightCurve_WASP-50 b_2013-12-16.png (b87d6f9f2e96...), FinalLightCurve_WASP-50 b_2013-12-16.csv (78f89d80261a...)

Compute resources

Wall time 245.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 10:18Z.